Samineni Sadhwika

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.12
NARAYANA JUNIOR COLLEGE, CURRENCY NAGAR, VIJAYAWADA	Class XII (SSC)	2024	97.50%
VISWABHARATI HIGH SCHOOL(EM) RAJENDRANAGAR, GVD KRISHNA DISTRICT	Class X (SSC)	2022	93.50%

SKILLS

Languages: C++, JavaScript, TypeScript

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS

Tools: Git/GitHub, VS Code

PROJECTS

Autonomous Local-First Agentic AI Operating System

TypeScript, Python, Node.js, Playwright, LLM Tool-Calling, React

- Built an autonomous agentic platform capable of decomposing complex natural language objectives into verifiable execution graphs with self-correcting feedback loops.
- Engineered multi-step browser automation and external tool calling pipelines executing end-to-end task completion with error-recovery mechanisms.
- Designed event-driven stateful execution engines to manage memory context, session persistence, and multi-agent coordination.

Audio Intelligence & Edge AI Classifier

Python, PyTorch, Torchaudio, SciPy, Librosa, NumPy, Post-Training Quantization (PTQ)

- Engineered an end-to-end audio pipeline converting raw 16 kHz acoustics into Log-Mel spectrograms via Short-Time Fourier Transform (STFT, 512-pt FFT, 160-sample hop) and Mel filterbanks.
- Designed a lightweight Depthwise Separable CNN (EdgeDSCNN with 56K parameters) achieving 99% classification accuracy across diverse acoustic events and speech commands.
- Integrated SpecAugment (time/frequency masking), dynamic SNR noise injection, and energy-based Voice Activity Detection (VAD) to improve acoustic robustness in noisy ambient environments.
- Applied Post-Training Quantization (INT8 PTQ), compressing model footprint to 0.23 MB with ~21 ms CPU latency for resource-constrained edge devices and hearables.

AI Speech Denoising & Acoustic Noise Suppression Autoencoder

Python, PyTorch, Torchaudio, SoundFile, SciPy, STFT/iSTFT Signal Reconstruction

- Developed a deep learning speech enhancement system using a convolutional encoder-decoder to suppress non-stationary background noise (traffic, fan hum, ambient babble) from 16 kHz speech.
- Designed the neural model to predict an Ideal Ratio Mask (IRM) bounded in [0, 1] over magnitude spectrograms, isolating human speech formants while preserving vocal harmonics.
- Synthesized clean time-domain audio waveforms from predicted spectral magnitudes and noisy phase using Inverse Short-Time Fourier Transform (iSTFT), achieving up to +8.67 dB SNR improvement.
- Formulated automated evaluation pipelines measuring Signal-to-Noise Ratio (SNR) gains and spectral MSE loss across varied acoustic noise degradation profiles.

LEADERSHIP & EXTRACURRICULAR

- National Cadet Corps (NCC), Army Wing, B Certificate -- selected as **Parade Pilot** for Republic Day and Independence Day parades.
- Completed the **Annual Training Camp (ATC)** at Karimnagar Police Academy, structured training in drill, discipline, and leadership

RELAVANT COURSES

Computer Science Courses: Introduction to Programming

Mathematics Courses: Introduction to Statistics, Introduction to Probability, Differential Equations, Calculus

Additional Courses: Spoken Japanese Basics